

DEEPAK RATHORE

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SUMMARY

AI/ML Engineer with production experience building RAG pipelines and Agentic AI systems. Proven track record of improving system performance by 90% and scaling enterprise platforms for clients like Havells and TVS using LLMOps, Vertex AI, and AWS Bedrock.

TECHNICAL SKILLS

Languages: Python, SQL, Bash/Shell

ML & Deep Learning: Scikit-Learn, Keras/TensorFlow, Neural Networks, CNNs, Random Forests, Gradient Boosting, SVM, K-Means Clustering, Hyperparameter Tuning, Model Evaluation (F1/Precision/Recall/Confusion Matrix)

Generative AI & LLMs: LangChain (Chains, Agents, Memory, Document Loaders), LlamaIndex, OpenAI API, Gemini API, Hugging Face Transformers, LoRA/QLoRA Fine-Tuning, Prompt Engineering, Agentic AI / Multi-Agent Systems

RAG & Vector Databases: RAG Pipelines, Semantic Search, Embedding Pipelines, ChromaDB, Pinecone, Weaviate

Cloud & LLMops: Vertex AI (Gemini Pro, RAG, Fine-tuning), AWS Bedrock, LLMops, CI/CD for AI (GitHub Actions/AWS), Streamlit Cloud, Docker, Heroku

Backend & Deployment: FastAPI, Django, RESTful APIs, Async I/O Architecture, Git/GitHub

Data & Document Processing: Pandas, NumPy, Matplotlib, Seaborn, PyMuPDF, Tesseract OCR

WORK EXPERIENCE

Softude Infotech Pvt. Ltd.

Jun 2025 – Present

AI/ML Engineer

Indore, India

- Architected a production LLM-driven conversational automation system for DCGPAC — integrating intent classification, RAG-based context retrieval, and multi-turn dialogue management to automate purchase orders, approval workflows, inventory queries, and vendor communication through a single chatbot interface, eliminating manual touchpoints across 3 enterprise workflows.
- Engineered event-driven workflow automation pipelines (webhooks, async task schedulers, background workers) that compressed multi-step operational processes from hours to under 5 minutes, removing human bottlenecks across procurement, logistics coordination, and document routing.
- Built a Maritime OCR Intelligence System processing port circulars, shipping notices, and regulatory PDFs using a multi-stage NLP pipeline (PyMuPDF + Tesseract + Transformer re-ranking + keyword scoring) — automatically surfacing profit-impacting insights that previously required hours of manual analyst review.
- Resolved critical architectural failures in the flagship Cost Estimation Platform serving Havells, TVS, and Hero — eliminated memory leaks, restructured N+1 queries into optimised joins, and migrated the data access layer to a fully async/await I/O architecture, achieving a 90% overall performance improvement.
- Scaled stable concurrent user capacity from 3–4 to 25–30+ users per client and reduced peak API response time by 70% through systematic profiling, connection pooling, and async I/O refactoring across the full backend stack.

RA Software

Feb 2024 – Aug 2024

Backend Developer Intern

Jaipur, India (Remote)

- Built and maintained RESTful APIs, improving response consistency across 5+ endpoints handling 10,000+ daily transactions. Designed normalised database schemas and optimised query logic (index tuning, query batching).
- Contributed to Agile sprint cycles — sprint planning, code reviews, and CI/CD pipeline maintenance — gaining hands-on production debugging and deployment experience.

PROJECTS

OMNIMIND — Full-Stack Agentic AI Research Platform *Python · LangChain · LlamaIndex · FastAPI · ChromaDB · Docker · Streamlit*

2026

- Built a multi-agent research platform with three execution paths — a LlamaIndex + ChromaDB RAG pipeline, a LangChain + Tavily web-search ReAct agent, and a GPT-4o reasoning agent — fronted by FastAPI and a Streamlit UI.
- Designed a query router that dispatches each question based on live-keyword detection, query-complexity heuristics, and top-K retrieval confidence from a FlashRank-reranked ChromaDB lookup, with forced-route overrides for evaluation.
- Implemented an OCR ingestion pipeline (PyMuPDF + Tesseract fallback) feeding a MiniLM embedder, with batch upload, document listing, and deletion APIs.
- Wrote QLoRA (4-bit) fine-tuning and MRR@10 / NDCG@10 / Hit@5 evaluation scripts for the retriever embedding model.

EDUCATION

Rajiv Gandhi Proudyogiki Vishwavidyalaya (RGPV), Bhopal

March 2025

B.E. — Computer Science & Engineering (7.6 CGPA)